



Indian Institute of Information Technology Vadodara

Siddhikesh Arvind Gavit

Education

Indian Institute of Information Technology Vadodara

B.Tech in Computer Science and Engineering **CPI: 6.33**

Aug 2023 – May 2027

Gandhinagar, Gujarat, India

Hon. Sharad Pawar International School

Class XII (Higher Secondary) – CBSE **Percentage: 80.00%**

June 2020 – Mar 2022

Nashik, Maharashtra, India

Hon. Sharad Pawar International School

Class X (Secondary) – CBSE **Percentage: 84.80%**

June 2011 – Mar 2020

Nashik, Maharashtra, India

Experience

IIIT Vadodara

June 2026 – August 2026

Research Intern – Edge-AI / On-Chip Implementation of Sleep-Quality Prediction

Research Internship, Gandhinagar

- Researching an **on-chip (edge-hardware) implementation** of a sleep-quality prediction model for **real-time, low-power on-device inference** from wearable biosignals.
- Built and benchmarked **7+ ML/DL models** (LSTM, GRU, TCN, Transformer, XGBoost) with a **soft-voting ensemble** to predict next-day **WASO** from 7-day wearable **HRV** and questionnaire biomarkers, reaching **AUROC ~0.87**.
- Ensured leakage-free evaluation with **subject-wise splits**, plus **SHAP + LIME** explainability and Youden's-J threshold tuning (**Python, TensorFlow, scikit-learn**).

Projects

Multi-Agent AI Research Assistant | *Python, LangGraph, LangChain, LLM APIs, ChromaDB, Tool-Calling, Streamlit*

- Built an **autonomous multi-agent system** that decomposes a query, retrieves evidence from web and documents, and synthesizes a **cited, structured report** grounded in the sources.
- Orchestrated specialized agents (**planner, researcher, writer, critic**) in **LangGraph** with tool-calling, shared vector memory, and iteration limits to prevent runaway loops.
- Evaluated outputs on a **20-query test set** using **RAGAS** (groundedness & citation precision), raising groundedness to **~0.90** and reducing unsupported claims versus a single-prompt baseline.

AI Workflow Automation Platform | *Python, n8n, LangChain, LLM APIs, Docker, REST APIs, Webhooks*

- Designed and deployed an **end-to-end automation** that ingests emails and form submissions, classifies intent with an **LLM**, and auto-routes structured responses across Gmail, Sheets, and Slack – cutting manual triage effort by **~70%**.
- Hardened the pipeline with **webhook retries, idempotency, and failure alerts**, containerized via **Docker** for reliable scheduled and asynchronous runs.
- Engineered a few-shot **prompt layer** to extract clean JSON from unstructured text, with logging and run-monitoring to track success rate and surface failures.

Production-Ready ML Prediction Service | *Python, scikit-learn, XGBoost, Pandas, FastAPI, Docker*

- Developed and deployed a **churn-prediction model** as a **REST microservice** (FastAPI + Docker), covering the full pipeline: **EDA → features → training → deployment**.
- Improved **F1 from 0.72 to 0.85** on a **7K+** record dataset by handling class imbalance with **SMOTE** and tuning **XGBoost** via GridSearchCV.
- Served **real-time predictions** through a containerized API with input validation, logging, and **sub-100 ms** latency for easy integration into downstream apps.

Achievements

Anveshan Hackathon, IIT Guwahati (national-level) – built an ML model for **Fake Job Scam Detection** under a 48-hour deadline.

Participant – **Adobe Hackathon** and **2× HackIIITV** (institute flagship), building working prototypes in 24–36 hours.

Football (State Level) – **3×** state-level player representing **Nashik Division, Maharashtra**; appeared in **SGFI National** trials.

Football (College) – **2nd place, Concours - DAIICT Sports Festival**; **3rd place, national Inter-IIIT Sports Meet** (representing IIIT Vadodara); **2× 2nd place, Ventura** (2024 & 2025).

Technical Skills

Languages: Python, SQL, Java

Generative AI & Automation: LLM APIs (OpenAI, Gemini, Claude), LangChain, LangGraph, LangSmith, Prompt Engineering, RAG, Vector Databases (ChromaDB, Pinecone), Hugging Face, n8n, Zapier, Make.com

Data Science & ML: Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, EDA, Feature Engineering, Model Evaluation, Jupyter

Frameworks & Web: Flask, FastAPI, Streamlit, REST APIs

Databases & Tools: MySQL, PostgreSQL, MongoDB, Supabase, Git, GitHub, Docker, Postman, VS Code, Google Colab

Core CS: OOP, DBMS, Operating Systems, Computer Networks, Data Structures & Algorithms

Volunteering/Administrative Role/Managerial Role

Teaching Assistant & Football Mentor

Aug 2025 – Present

Mentor for First-Year Students

IIIT Vadodara, Gandhinagar

- Appointed as a **paid Teaching Assistant** to mentor and coach first-year students in football – led practice drills and guided match preparation.

Obscura – Photography Club

2023 – Present

Member & Event Photographer

IIIT Vadodara, Gandhinagar

- Photographed flagship campus events – **Kreiva** (annual cultural fest), **Ventura** (sports fest), **Freshers' Party**, and **HackIIITV** (hackathon).